

Acids and Bases

Chemical 18 Equations

Making Acids

Acids are usually formed when gases made from non-metallic elements are dissolved in water.

Task 1

Match each gas with the acid it can produce when dissolved.

Gas dissolved in water	Acid formed
Hydrogen chloride	Nitric acid
Sulfur dioxide	Carbonic acid
Carbon dioxide	Hydrochloric acid
Nitrogen dioxide	Sulfuric acid

Task 2

Acid rain is actually a weak solution of sulfuric acid. Acid rain is caused by the burning of fossil fuels. Write word equations (with balanced symbol equations if you are able) for each step of this process.

Step 1 - Sulfur impurities in the fossil fuel burn to form sulfur dioxide. This gas escapes into the air.

Step 2 - The sulfur dioxide is slowly oxidized in the atmosphere to form sulfur trioxide.

Step 3 - The sulfur trioxide dissolves in the rainwater to form sulfuric acid.

Making Bases

Bases can be made by dissolving very reactive metals in water.

Task 3

Write word equations (and balanced symbol equations if you can) showing how each of the metals below reacts with water to produce the base shown. Each reaction also produces **hydrogen gas**.

	Reactant dissolved in water	Base formed
a.	Potassium	Potassium hydroxide
b.	Sodium	Sodium hydroxide
c.	Calcium	Calcium hydroxide

Task 4

Oxides of the reactive metals will also dissolve in water to produce bases. Write a word equation to show how calcium oxide reacts with water to produce calcium hydroxide.

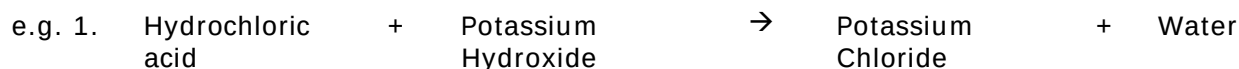
Neutralization Reactions (Acids and Bases continued)

Acids will react with bases to form a neutral solution and water. These are called neutralization reactions.



Remember:

1. If hydrochloric acid is used, then the salt is a CHLORIDE
2. If sulfuric acid is used, then the salt is a SULFATE
3. If nitric acid is used, then the salt is a NITRATE



Task 5

Write word equations for the reactions between the following pairs of reactants. If you have been taught how to, then write a balanced symbol equation under each word equation. Remember to set out your word equations in the manner shown above.

	Reactants	Products (fill first if helpful)
a.	Hydrochloric acid and sodium hydroxide	
b.	Sulfuric acid and potassium hydroxide	
c.	Nitric acid and calcium hydroxide	
d.	Hydrochloric acid and calcium hydroxide	
e.	Nitric acid and potassium hydroxide	
f.	Sulfuric acid and calcium hydroxide	
g.	Hydrochloric acid and lithium hydroxide	
h.	Nitric acid and sodium hydroxide	
i.	Nitric acid and lithium hydroxide	
j.	Sulfuric acid and lithium hydroxide	

Task 6

Write word equations (and symbol equations if possible) for the neutralization reactions below.

- a. A medicine containing weak magnesium hydroxide can be used to treat the indigestion caused by excess hydrochloric acid in the stomach.
- b. Slaked lime (calcium hydroxide) can be used to neutralize dangerous hydrochloric acid spills.